Graph Algorithms & Shortest Paths Notes

Dijkstra's Algorithm:
Dijkstra's algorithm finds the shortest path from a single source vertex to all other vertices in a weighted directed or undirected graph with non-negative edge weights.

Priority Queue Implementation:
Using a min-priority queue (binary heap), the time complexity is O((V + E) log V). In each iteration, the algorithm extracts the vertex u with the smallest tentative distance and relaxes all outgoing edges (u, v, w).

Relaxation condition:

```
if dist[u] + weight(u, v) < dist[v]:
    dist[v] = dist[u] + weight(u, v)
    pq.push((dist[v], v))
```

Key Properties:
- Greedy approach: always processes the closest unvisited vertex.
- Non-negative edge weights constraint: Dijkstra fails with negative weights; use Bellman-Ford instead.
- Priority queue maintains the frontier of reachable vertices.